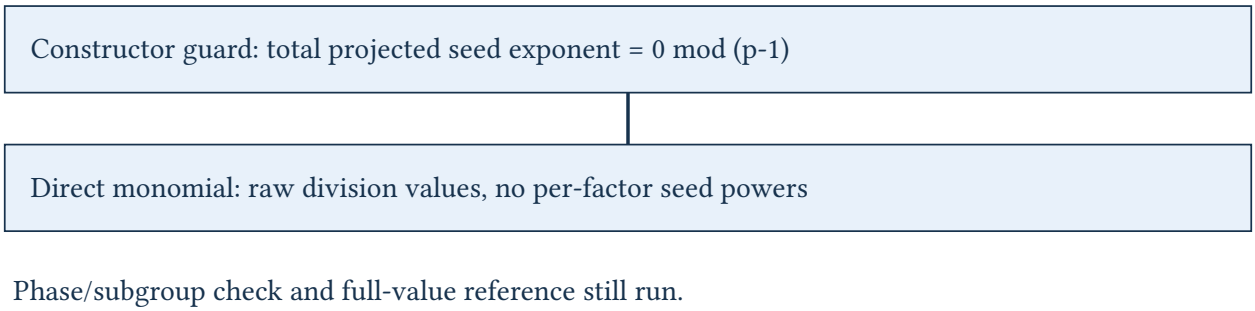
Projected monomial seed cancellation

2026-11-07 | PCC-20261107-c45d3c / PCC-20261107-c64866



Derived construction. The combined seed exponent is divisible by four in both partition-parity cases. For image order two or four, raising to the odd part of p-1 therefore cancels the seed. Direct monomial initialization uses raw division values. A constructor guard checks modular exponent closure. [pcc:unverified]

Observed development checks. Four selected native methods pass, including closure on 3479 algebraic partitions and rejection of 3479 changed seed exponents. Existing paired-source agreement covers 499 subgroup rows, 100 splits, 1375 advances and 450 skips; limits and factor-request checks also pass. These are unsigned producer results. The algebraic cases are not a curve cohort. [pcc:unverified]

Scope and cost. Phase/subgroup calculation and the full-value reference remain. This removes per-factor seed powers and twice-base seed construction, not all phase or residual work. No scalar recovery, comparative timing or cloud job ran. Original cohort intact.

Failure retained. First compile failed on an ambiguous integer type in the new test. The initial-build snapshot preserves it. A u64 annotation fixed compilation; no native test failed.

Next. Replace unused projected-path phase-value computation with sufficient subgroup validation. Investigate projected residual work, then complete-cost accounting.

Full parity proof: DERIVATION.md. Exact commands/hashes: receipt.json and source-lock.json. Sources: predecessor factor/projection derivations and phase implementation, with the previously reviewed Sutherland cyclic-group and Sage division-polynomial references. KB unavailable. No novelty-priority claim.